

7 ways to find the queen

(And one way to get a result without finding the queen)

1 Go the Centre

- [a] Use minimal smoke, 1-2 puffs, take off honey super and queen excluder.
- [b] Space frames apart in the middle of the brood box.
- [c] Scrape off burr comb on side of top bar
- [d] Carefully lift one frame out of the middle taking care not to squash bees.
- [e] Briefly scan over each side of this frame (like reading, taking about 1 minute). Place this frame in a spare box and continue with the next frame.

The queen can be on the wall of the hive body or on the bottom board. The centre of the brood nest on a frame with eggs is the most likely place to find her. Speed is an advantage because the old queen can hide well. Have an empty cage handy, if you find her on the first frame you might recycle her into a drone layer or an egg less hive later if she has a good brood pattern or has other good qualities. Once you've found her; shake bees off combs, check for disease, manipulate combs, scratch honey in corners to make egg laying space, lift honey out to top and replace with empty brood combs, introduce cage and mark the front of box with code - date - breeder. W123

2 Go the Outside

Otherwise known as the light barrier or Linton Briggs method.

- [a] Place an empty box next to the hive.
- [b] Use minimal smoke, 1-2 puffs, take off honey super and queen excluder.
- [c] With the sun over your left shoulder, remove the outside frame furthest from you - checking for queen before placing in the empty box.
- [d] Remove outside frame closest to you, again checking for the queen before placing in the spare box. While it is not usual to find the queen on the outside frames (it can happen), by removing these frames first you create a light barrier between the next frame and hive wall. This will confine the queen to the remaining frames.
- [e] Before checking both sides of the next frame closest to you, glance down the face of the frame. According to Linton the queen often stands taller than the rest of the bees, and can be spotted up to 20% of the time on the face of the frame before it is removed. Repeat for remaining frames until the queen is found.

3 Divide and Conquer

- [a] Put an empty box next to the hive.
 - [b] Take out half the frames and place in the empty box. Place in each box empty combs.
 - [c] You will then have 4 frames of bees and brood, and 4 frames of pollen/honey/empty combs in each box. The next day one of the hives will have fanning bees at the front entrance (looking to attract the queen). The queen will be in the hive with no fanning bees at the entrance. She will now be easier to find because you know which box she is in and have only half the bees to search.
- If you have 2 brood boxes, by splitting them you get a similar result - half the bees with the queen in one box. The beauty of this method is it only takes seconds to do.

4 Last Resort

This one is as a last resort and perfect for a recycled queen, and is suitable to use on a drone layer or stinging hive.

- [a] You move the whole hive 10-20 meters or more, behind landmarks preferably. This will cause them to drift back to their original position.
- [b] Shake all bees off combs onto ground and place all frames into a new hive body which is elevated on a table. This ensures that the queen cannot re-enter the hive.
- [c] Replace this brood box onto original spot to collect all field bees.
- [d] The queen will not be able to fly back.
- [e] Check 7 days later for queen cells, knock them off and introduce cage queen or cell.

5 Drift Method

Suitable for pallets or apiaries with pairs or rows of hives.

- [a] Move hive to a new position a few meters away behind a landmark and turn entrance 180 degrees.
- [b] This will cause all the field bees to return to the hive next-door. Usually this is OK under good conditions.
- [c] The next day or even a few hours later you only have nurse bees and the queen on the brood frames; this will give us the edge in finding the old queen.

6 Divide and Divide Again

- [a] Place an empty hive body next to hive.
- [b] Use method [1] first.
- [c] Place four frames in each box with bees adhering. Pair up the frames. This should make the bees and the queen go between either two pairs of frames where it is darker. After 10 minutes look for the queen and remember to look on the wall and floor as well.

7 Strainer Method

This one will demoralise the hive for a while, so should only be used if all else fails.

- [a] Move hive onto next hive and replace an empty body on that original place. On top of that we place a super with the queen excluder screwed on the bottom (strainer box).
- [b] We shake all bees into this strainer box and then return the brood frames from the original box under this strainer into the brood box on its original location.
- [c] With a little smoke, force bees through strainer back into brood.
- [d] You will only have drones and the queen left in this strainer - use only cool smoke, so the bees don't panic and run up the walls of strainer.

A slight variance of this version is the Bob McDonald method. Move the hive and replace a new bottom board, put a queen excluder on the bottom then place empty box, shake all bees in front of original position placing brood frames into this box. The bees will enter moving up through the queen excluder to look after brood. When the hive has settled down, lift brood box off the excluder and the queen will be trapped between the excluder and the bottom board. Place this bottom board and excluder in a hessian bag and remove.

8 Supersedure Method (Autumn only)

This method involves splitting the hive to introduce queen cells, possibly leaving the old queen.

- [a] Shake all bees off brood combs into old brood box and place into new super (not to be done under heavy honey flow because you risk drowning bees in their nectar).
- [b] Fill brood box with empty brood combs, replace queen excluder then place box of brood on top. Nurse bees will move up through queen excluder to feed the young. This can be done up to 7 days before.
- [c] The old queen will occupy empty combs and start laying there. After 6 hours or next day the hive can be split with a division board or taken away to new site. Either way you introduce the new queen into this brood box. Since the old queen has lost brood and bees she should be easier to find in the next few days.

By introducing a queen cell placed in a cell protector or (hair roller) it should survive and hatch the next day. When the virgin queen hatches it does not have any queen pheromones and the old queen is used to bees walking over her and grooming her therefore the chances of the old queen being killed by the virgin queen with a sting is very high. This method is widely used by commercial beekeepers overseas as it is very quick and should give us 80% success, so I would split stronger hives to give us over 100% of laying young queens for the coming season. For this to work, we need good drone population and a breeding flow [nectar & pollen].